Lab 10 - AFSK

Objectives

- Test the FSK receiver you coded in the previous lab on an AFSK packet transmitted via a radio link.

Equipment needed

- Computer running Matlab.

Background

AFSK stands for Audio Frequency Shift Keying. It is FSK modulation using the audio channel of a link. This can be used to implement a modem over a phone line or a radio link. AFSK is commonly used by amateur radio operators to transmit digital data using standard FM radios.

Procedure and Analysis

- On the course website is a file named “newwav.wav” that is a recording of an afsk signal sent over a radio link. The signal consists of the same sync pattern we used in the previous lab and then some random data bits. The sampling rate is 4000 samples per second. The bit period is 0.1 second. The center frequency is 500 Hz and the frequency deviation is 10 Hz. To read this file into a MATLAB row vector *r*, give the command: [r,fs] = audioread(‘newwav.wav’);
- Use the Matlab code that you developed and tested in the previous FSK lab to determine the data bits represented in the file newwav.wav.

Write Up

- Present the received data bits as 1’s and 0’s.
- Present plots that demonstrate the operation of your receiver.
- Discuss your receiver’s performance.